



Follow the Steps

Run the algorithm to make a pattern

Name: _____

Date: _____

★ I can follow the steps of an algorithm to make a pattern.

Run the algorithm

Follow the steps in order, one at a time, to fill the boxes. Each step tells you which shape to draw in the next box. Running the steps makes a pattern.

The algorithm

1. box 1 → draw ●

2. box 2 → draw ■

3. box 3 → draw ●

4. box 4 → draw ■

5. box 5 → draw ●

6. box 6 → draw ■

Draw here — boxes 1 to 6



1 Follow the steps

Do each step in order and draw ● or ■ in the boxes above. Read box 1 first, then box 2, and so on.

2 Read your result

What shape is in box 6? Draw it or write its name.

3 You try — keep the pattern

What is the pattern called (● ■ ● ■ ...)? If there were a box 7, what shape would go there? Tell how you know.

Big idea

Running an algorithm means doing each step in order. The same steps always make the same result — here, a circle-square pattern.



Follow the Steps

Quick facilitation notes & answer key

What this worksheet teaches

Students follow a 6-step set of instructions in order to fill a row of boxes, making a circle-square pattern, then read and continue it. Following steps in order to build a result is running a plan. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: cut the steps into cards so students can physically slide them into order before writing.

How to lead it (about 20 minutes)

1. Show them (you do). Read box 1, draw its shape; read box 2, draw its shape — one box at a time, in order.
2. Do one together. Fill the first few boxes together (circle, square, circle...).
3. Let them try. Students finish the row, then read and continue the pattern.

Answer key (these are the verified correct answers)

Exercise 1 — boxes 1–6: circle, square, circle, square, circle, square. Exercise 2 — box 6 is a square.

Exercise 3 — the pattern is circle-square (an AB pattern); box 7 would be a circle, because the pattern repeats and box 6 ended on a square.

Why it works: each step runs in order and changes only its own box, so the row always comes out the same.

If students get stuck — and ways to adjust

Watch for: skipping a box or doing them out of order. Read the boxes strictly 1, 2, 3...

Make it easier: point to each box as you read its step.

Make it harder: ask what box 8 would be and how they know.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students put step-by-step instructions in the right order to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a set of steps, change or reorder one, and explain what the change did.

You'll know it worked when

The child fills the row in order and continues the pattern correctly.